

How few donor basins are enough?

Regional transfer learning for daily streamflow with two years of local data

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1 Introduction

Most gauges do not come with decades of discharge. The usual large-sample LSTM recipe—hundreds of basins, long sequences, a GPU—answers a different question from the one a field office actually faces: can a regional model help when the local record is about two years long and the hardware is ordinary? That constraint is the study design here.

The experiment holds the target, the warmup, and the evaluation window fixed, and varies only how the regional prior is built. An entity-aware LSTM (EA-LSTM) is pretrained on similar Midwest donors, then transferred to Lick Creek near Perry, Missouri (USGS 05507600) with a frozen-LSTM, head-only fine-tune and a walk-forward refit protocol. Training from scratch on the same warmup is the local baseline. Next-day persistence and a day-of-year climatology—fit only on that warmup—put a floor under the NSE numbers. A later sensitivity run cuts the donor pool to eight similar basins.

The numbers in `results/midwest_mini/` are from this completed Lick Creek study (1,461 evaluation days), not from a smoke test or synthetic sample. Code is at github.com/krishharma/hydro-tl-ews.

2 Data and methods

2.1 Basin, donors, and periods

CAMELS-US supplies Daymet weather, USGS streamflow, and 27 static catchment attributes. The target is a plains basin (snow fraction ≈ 0.07). Donors are the most similar Midwest gauges in static attribute space inside $[36.5^\circ\text{N}, 49.5^\circ\text{N}] \times [-104^\circ\text{W}, -80.5^\circ\text{W}]$, after a 50 km exclusion buffer and after dropping gauges missing from the local HUC 04/05/07/10 extract. That procedure yields **17 donors** (Figure 1). Weather inputs are precipitation, $T_{\max}$, $T_{\min}$, shortwave radiation, vapor pressure, and day length.

2.2 Model and transfer

The model is an EA-LSTM: static catchment attributes control the input gate; weather and memory drive the remaining gates. Hidden size is 128, lookback **90** days, dropout 0.4, forget-gate bias +3.0. Pretraining runs 10 epochs on the 17 donors. Adaptations compared: zero-shot; Approach A (output

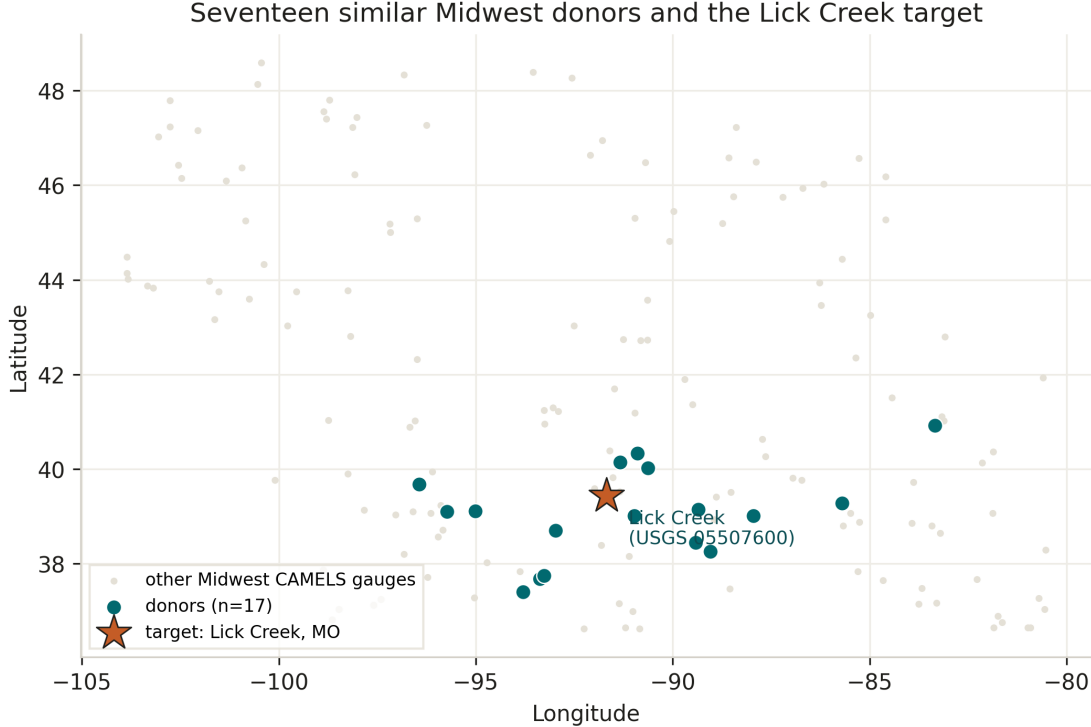


Figure 1: Seventeen similar Midwest donor gauges and the Lick Creek target. Grey points are other CAMELS gauges in the same box; they are context, not training data.

Table 1: Time periods used for USGS 05507600.

Period	Dates	Role
Donor pretrain	Oct. 1995–Sep. 2008	Train the regional EA-LSTM
Pretrain validation	Oct. 2008–Sep. 2010	Early stopping
Target warmup	Jan. 2009–Dec. 2010	Scarce local record (~ 2 yr)
Target evaluation	2011–2014	Fixed-window and walk-forward tests
Threshold history	1990–2010	At-site Q5, Q95, Q99

head only; LSTM frozen); local-from-scratch on the warmup; walk-forward with a head-only refit every 90 days from 2009-01-01 and online bias correction. Approach B was not run here. The scored window is 2011–2014 ($n = 1461$ days).

2.3 Naive baselines and warnings

Persistence sets $\hat{Q}(t) = Q(t - 1)$. Day-of-year climatology is the mean observed discharge on that calendar day during warmup 2009–2010 only. Warnings are a post-process on the walk-forward hydrograph. Labels at date t with lead L are 1 if any day in $[t + 1, t + L]$ crosses the threshold. Flood thresholds are $Q95 = 3.26 \text{ mm d}^{-1}$ and $Q99 = 16.91 \text{ mm d}^{-1}$. **Q5 resolves to 0.0 mm d^{-1}** , so those events are low/zero-flow days, not a seasonal drought. Original probabilities use a Gaussian residual and an independence product over the lead window. A second estimator takes the maximum of the 1-day probabilities in the same window. Metrics: NSE, KGE, PBIAS, AUC, Brier, BSS against the event base rate, F1 at 0.5 and at the threshold in $\{0.05, \dots, 0.95\}$ that maximises F1 on the scored

window (diagnostic only).

3 Results

3.1 Continuous streamflow skill

Table 2 places transfer above both naive baselines and the local LSTM. Walk-forward NSE (0.179) is the best continuous score. Persistence has a high KGE (0.42) because yesterday is correlated with today, but its NSE is negative (-0.166): peak errors on this flashy plains hydrograph more than eat the variance. Climatology from two years of warmup is worse ($\text{NSE} = -2.34$). Local-from-scratch barely beats a mean forecast ($\text{NSE} = 0.009$). Fine-tuning almost removes volume bias ($\text{PBIAS} +4.1\%$). Absolute NSE is modest next to continental LSTMs; the ranking that answers the study question is not: **transfer** $\gg$ **local**, and transfer beats persistence on NSE.

Table 2: Continuous skill at USGS 05507600, 2011–2014 ($n = 1461$ days).

Setting	NSE	KGE	PBIAS (%)
Next-day persistence	-0.166	0.421	+1.6
Day-of-year climatology (warmup)	-2.337	-1.486	+223.1
Local baseline (from-scratch warmup)	0.009	-0.351	+72.6
Zero-shot (pretrained only)	0.150	0.032	-37.5
Fine-tune, head only (Approach A)	0.160	0.112	+4.1
Walk-forward (17×90 -day refits)	0.179	0.161	-8.7

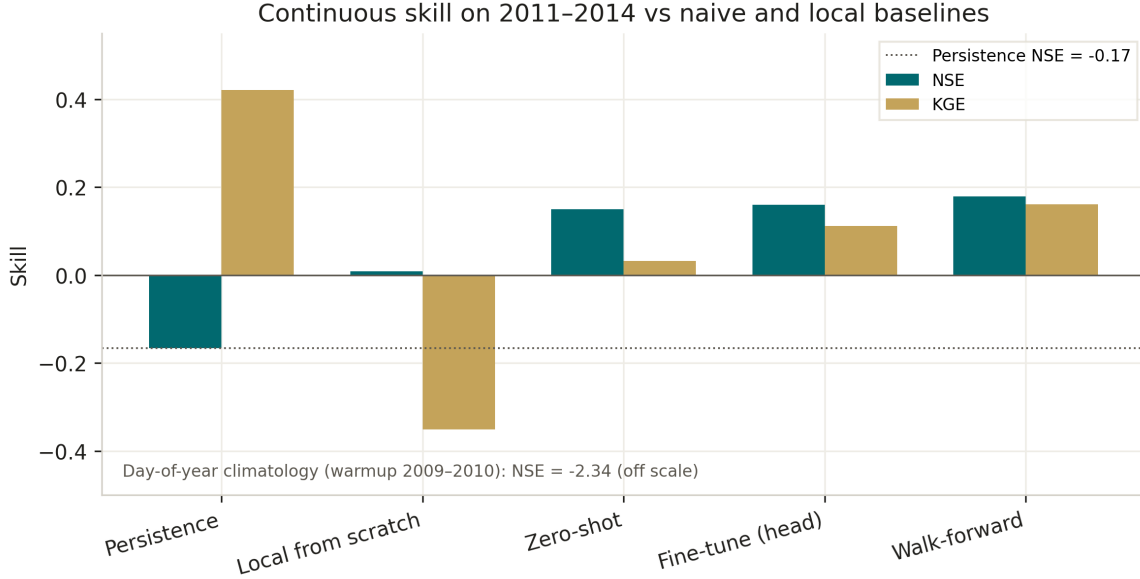


Figure 2: NSE and KGE across settings. Dotted line: persistence NSE. Day-of-year climatology ($\text{NSE} = -2.34$) is annotated, not forced onto the same axis.

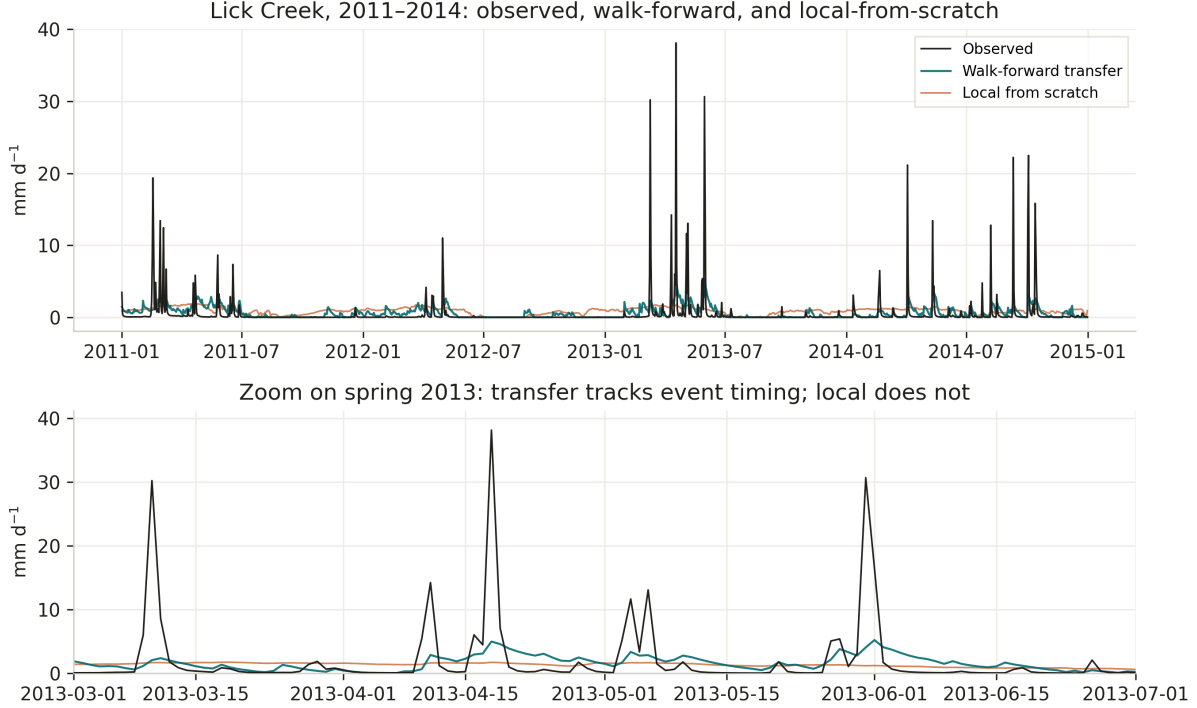


Figure 3: Observed discharge versus walk-forward transfer and local-from-scratch. Transfer follows event timing; the local model does not. Predictions clipped at zero for display.

3.2 Early-warning scores and calibration

Flood Q95 discrimination is strong at every lead (AUC 0.908–0.922). Raw Brier scores still look as if skill collapses from 0.094 (1-day) to 0.693 (7-day). BSS against the base rate is negative for the independence product: mean 1-day probability is 0.29 while the event rate is 0.034, and the 7-day product has mean probability 0.89. Ranking is intact; calibration is not. Replacing the product with a window-max of 1-day probabilities cuts 3-day Brier from 0.374 to 0.105 and 7-day Brier from 0.693 to 0.121. F1 at 0.5 is the wrong operating point; a threshold sweep raises 1-day F1 from 0.286 to 0.452 at probability 0.45 (Table 3; Figure 4).

Table 3: Walk-forward warning scores. Prod. = independence product; max = window-max of 1-day probabilities. F1* is the best F1 on a 0.05–0.95 sweep (diagnostic). Q5 = 0 mm d^{−1} (low/zero flow).

Event	Lead	AUC	Brier _p	BSS _p	Brier _m	BSS _m	F1@0.5	F1*
Flood Q95	1 d	0.922	0.094	−1.90	0.094	−1.90	0.286	0.452
Flood Q95	3 d	0.914	0.374	−4.87	0.105	−0.65	0.133	0.500
Flood Q95	7 d	0.908	0.693	−5.45	0.121	−0.13	0.219	0.621
Flood Q99	1 d	0.950	0.005	—	0.005	—	—	—
Low/zero flow (Q5)	1 d	0.805	0.214	−0.74	0.214	−0.74	0.410	0.419

3.3 SHAP drivers

GradientExplainer ranks precipitation first, then vapor pressure, shortwave radiation, day length, $T_{\min}$, and $T_{\max}$ (Table 4). Static attributes have mean $|\text{SHAP}| = 0$ in this single-basin explanation

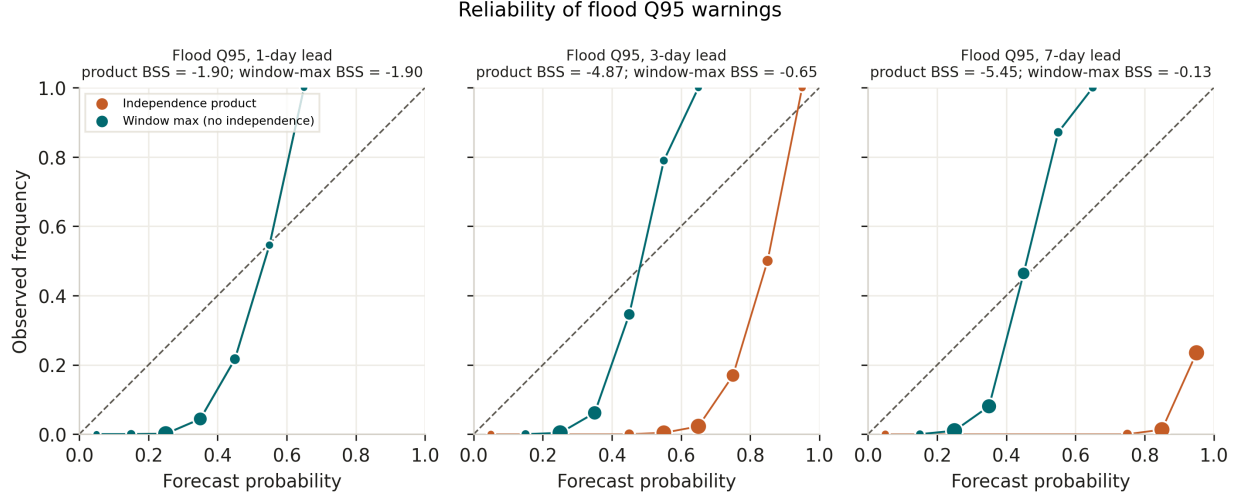


Figure 4: Reliability of flood Q95 warnings. Coral: independence product. Teal: window-max of 1-day probabilities.

because they do not vary within one gauge.

Table 4: Top SHAP drivers for USGS 05507600.

Feature	Mean SHAP
Precipitation (mm/day)	0.0097
Vapor pressure (Pa)	0.0055
Shortwave radiation (W/m ²)	0.0052
Day length (s)	0.0041
$T_{\min}$ (°C)	0.0036
$T_{\max}$ (°C)	0.0035
Static catchment attributes (all)	0.0000

3.4 How few donors? A negative result

Table 5 keeps the donor-pruning comparison. Cutting 17 similar donors to 8, with $T_{\min}/T_{\max}$ unchanged, lowers walk-forward NSE from 0.179 to 0.138. Combining pruning with mean daily temperature and a three-seed average recovers some of that loss (NSE 0.165) but still does not beat the 17-donor run. **The third row of Table 5 changes three variables at once** (donor count, temperature features, and multi-seed averaging) and should not be read as a single-factor effect. Forget-gate bias +3.0 is already the default in every row. Transfer still beats local-from-scratch after pruning.

Table 5: Sensitivity to donor pruning (USGS 05507600, 2011–2014 NSE). The third row changes donor count, temperature inputs, and seed averaging together.

Setting	Donors	FT	Local	WF
$T_{\min}+T_{\max}$, seed 42	17	0.160	0.009	0.179
$T_{\min}+T_{\max}$, seed 42	8	0.113	0.009	0.138
Mean T , 3-seed average	8	0.133	-0.097	0.165

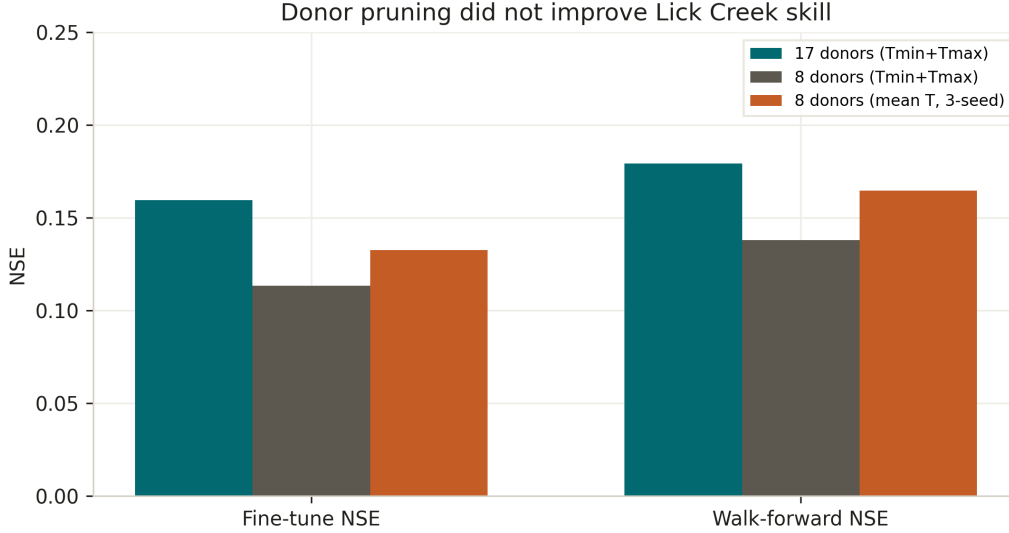


Figure 5: Seventeen similar donors outperform an 8-donor pool on both fine-tune and walk-forward NSE.

4 Discussion and conclusion

This is a one-basin Midwest study on commodity hardware. It does not estimate continental CAMELS skill or snowmelt-regime transfer. What it does estimate, with a full 2011–2014 evaluation, is whether a small similar-donor prior beats local training and naive baselines when local data are scarce. It does.

Warnings are hindcast post-processing of daily flow. F1* uses the same window it scores. Daily 1-day probabilities remain overconfident ($BSS = -1.90$) even before compounding; recalibrating that 1-day probability is future work.

With two years of local discharge and a laptop, seventeen similar Midwest donors are enough for regional transfer to beat both a local LSTM and next-day persistence on NSE. Eight donors are worse, not better. Flood Q95 warnings discriminate well; the ugly raw Brier numbers at long leads are mostly an independence assumption, not a collapse of ranking skill.

Midwest run: `python scripts/run_midwest_mini.py`. Baselines and figures: `python scripts/analysis/poster_from_csvs.py`.

Word manuscript: `python scripts/paper/build_short_paper_docx.py`.